

① $P_0 + S_0 = C_0 + ke^{-rT}$

$$P_T + S_T - C_T \begin{cases} S_T > k & 0 + S_T - (S_T - k) = k \\ S_T \leq k & k - S_T + S_T - 0 = k \end{cases}$$

$$P_T + S_T - C_T = k \quad \checkmark$$

$$\Rightarrow P_0 + S_0 - C_0 = ke^{-rT}$$

$$P_0 + S_0 = C_0 + ke^{-rT}$$

② Lower bound

$$P_0 + S_0 = C_0 + ke^{-rT}$$

$$\Rightarrow C_0 = P_0 + S_0 - ke^{-rT}$$

$$\geq S_0 - ke^{-rT} \quad \checkmark$$

$$C_0 \geq \max(0, S_0 - ke^{-rT})$$

$$P_0 = \underline{C_0} + ke^{-rT} - S_0$$

$$\geq \max(0, ke^{-rT} - S_0)$$

upper bound

Proof by contradiction.

X assume $C_0 > S_0$ ~~✗~~

$\rightarrow \pi_0 = S_0 - C_0 < 0$ (profit)

$\rightarrow \pi_T = S_T - C_T \geq 0$

no arbitrage. $\rightarrow C_0 \leq S_0$

$$C_0 = \mathbb{E}^Q(e^{-rT} \cdot \max(S_T - k, 0))$$

$$\leq \mathbb{E}^Q(e^{-rT} \cdot S_T)$$

$$= e^{-rT} \mathbb{E}^Q(S_T)$$

$$= e^{-rT} \cdot S_0 e^{rT} = S_0$$

$$C_0 \leq S_0 \quad \checkmark$$

$$P_0 = \mathbb{E}^Q(e^{-rT} \cdot \max(k - S_T, 0))$$

$$\begin{aligned}
 P_0 &= \mathbb{E} \left(e^{-rT} \max(k - S_T, 0) \right) \\
 &\leq \mathbb{E}^Q (e^{-rT} \cdot k) \\
 &= k e^{-rT}
 \end{aligned}$$

$$P_0 \leq k e^{-rT}$$

$$\begin{bmatrix} 0, (-) \\ T, \geq 0 \end{bmatrix}, \underline{\underline{0}}, \underline{\underline{\geq 0}}$$

③ Given $C_0(S_0, k, T_1) < C_0(S_0, k, T_2)$, $T_2 > T_1$

Assume $C_0(S_0, k, T_2) < C_0(S_0, k, T_1) \times$

$$\pi_0 = C_0(S_0, k, T_2) - C_0(S_0, k, T_1)$$

$$\pi_{T_1} = \begin{cases} S_{T_1} > k & C_0(S_{T_1}, k, T_2) - (S_{T_1} - k) \geq 0 \\ S_{T_1} \leq k & C_0(S_{T_1}, k, T_2) - 0 > 0 \end{cases}$$

$$\pi_{T_1} \geq 0$$

④ Given $C_0(S_0, k_1, T) > C_0(S_0, k_2, T)$, $k_2 > k_1$

Assume $C_0(S_0, k_1, T) < C_0(S_0, k_2, T)$

$$\pi_0 = C_0(S_0, k_1, T) - C_0(S_0, k_2, T) < 0$$

$$\pi_T = \begin{cases} S_T \leq k_1 & 0 - 0 = 0 \\ k_1 < S_T \leq k_2 & S_T - k_1 - 0 = S_T - k_1 > 0 \\ S_T > k_2 & (S_T - k_1) - (S_T - k_2) = k_2 - k_1 > 0 \end{cases}$$

$$\pi_T \geq 0$$

$$\max(k - S_T, 0) \quad , \quad \begin{matrix} \max(k_1 - S_T, 0) \\ \max(k_2 - S_T, 0) \end{matrix}$$

$k_2 > k_1$

$$(5) \quad C_0(S_0, k_1, T) = P_0(S_0, k_1, T) + S_0 - k_1 e^{-rT}$$

$$\hookrightarrow C_0(S_0, k_2, T) = P_0(S_0, k_2, T) + S_0 - k_2 e^{-rT}$$

$$C_0(S_0, k_1, T) - C_0(S_0, k_2, T) = \underbrace{P_0(S_0, k_1, T)}_{\substack{\uparrow \\ \max(k_1 - S_T, 0)}} - \underbrace{P_0(S_0, k_2, T)}_{\substack{\uparrow \\ \max(k_2 - S_T, 0)}} + (k_2 - k_1) e^{-rT}$$

$$\leq (k_2 - k_1) e^{-rT}$$

$$\textcircled{6} \quad d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$\left(\frac{0}{0}\right) \rightarrow \frac{0}{\infty}$$

$$C_0 = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$P_0 = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$



$$\textcircled{T=0}$$

$$T \rightarrow 0$$

$$d_1 = d_2 = \begin{cases} +\infty, & S_0 > K \\ 0, & S_0 = K \\ -\infty, & S_0 < K \end{cases}$$

$$\Rightarrow N(d_1) = N(d_2) = \begin{cases} 1, & S_0 > K \\ 1/2, & S_0 = K \\ 0, & S_0 < K \end{cases}$$

$$C_0 = \begin{cases} S_0 - Ke^{-rT} = \underline{S_0 - K} \quad (\text{intrinsic value}) \\ (S_0 - K)/2 = 0 \\ 0 \end{cases}$$

$$C_0 = \max(0, \underline{S_0 - K})$$

$$\textcircled{7} \quad \underline{d_1} = \frac{\ln(S_0/K) + rT}{\sigma\sqrt{T}} + \frac{1}{2}\sigma\sqrt{T}$$

$$\underline{d_2} = \frac{\ln(S_0/K) + rT}{\sigma\sqrt{T}} - \frac{1}{2}\sigma\sqrt{T}$$

$$\sigma \rightarrow 0 \quad d_1 = d_2 = \begin{cases} +\infty & , S_0 > Ke^{-rT} \\ 0 & , S_0 = Ke^{-rT} \\ -\infty & , S_0 < Ke^{-rT} \end{cases}$$

$$N(d_1) = N(d_2) = \begin{cases} 1 & , S_0 > Ke^{-rT} \\ 1/2 & , S_0 = Ke^{-rT} \\ 0 & , S_0 < Ke^{-rT} \end{cases}$$

$$C_0 = \begin{cases} S_0 - Ke^{-rT}, & S_0 > Ke^{-rT} \\ 0, & S_0 \leq Ke^{-rT} \end{cases}$$

$$C_0 = \max(0, \underline{S_0 - Ke^{-rT}})$$

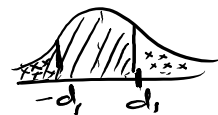
$$C_0 = S_0 N(d_1) - Ke^{-rT} N(d_2)$$

$$P_0 = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$$

$$\ln(S_0/K) + rT > 0$$

$$\Rightarrow \ln(S_0/K) > -rT$$

$$S_0 > Ke^{-rT}$$



⑧ $\sigma \rightarrow \infty$, $d_1 = +\infty$, $N(d_1) = 1$
 $d_2 = -\infty$, $N(d_2) = 0$

$N(d_1) = 1 - N(d_2)$

$$C_0 = S_0(1) - ke^{-rT} \cdot (0) = S_0$$

$$P_0 = ke^{-rT} \cdot (1) - S_0(0) = ke^{-rT}$$

⑨ Invariance

$$V(\alpha S, k, T) = \boxed{\alpha} \cdot V(S, k/\alpha, T) \quad \alpha > 0$$

$$\begin{aligned} V &= \mathbb{E}^Q \left(\max(\alpha S - k, 0) \right) e^{-rT} \\ &= \alpha \mathbb{E}^Q \left(\max(S - k/\alpha, 0) \right) e^{-rT} \\ &= \alpha \cdot V(S, k/\alpha, T) \end{aligned}$$

$$\frac{V(2S, k, T)}{V(S, k/2, T)} = 2$$

$$\textcircled{10} \quad v(\alpha S, \alpha k, T) = \alpha V(S, k, T)$$

d_1, d_2 are not impacted.

$$\begin{aligned} C_0 &= \alpha \left(S_0 N(d_1) - k e^{-rT} N(d_2) \right) \\ &= \alpha \cdot C_0 \end{aligned}$$

$$\underline{d_1} \leftarrow (-d_1)$$

$$\textcircled{11} \quad c(S, k, r, T, \sigma) = P(-S, \underline{-k}, r, T, \underline{-\sigma})$$

$$\begin{aligned} P &= \underline{k e^{-rT} N(-d_2)} - \underline{S_0 N(-d_1)} \\ &= S_0 N(d_1) - k e^{-rT} N(d_2) = c \end{aligned}$$

⑫ $V_T = \max(S_T^2 - k, 0)$

$$S_T = S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma W_T}$$

$$S_T^2 = \boxed{S_0^2} e^{\underline{2(r - \frac{1}{2}\sigma^2)T} + \boxed{2\sigma} W_T}$$

GBM ??

$$S_0 \rightarrow \boxed{S_0^2}$$

$$\sigma \rightarrow \boxed{2\sigma}$$

$$r \rightarrow \underline{x = (2r + \sigma^2)}$$

$$\max(S_T(S_T - k), 0)$$

$$\max(S_T^2 - S_T \cdot k, 0)$$

Stoak

$$\left(x - \frac{1}{2}(2\sigma)^2\right)T = 2(r - \frac{1}{2}\sigma^2)T$$

$$\Rightarrow x = 2r - \sigma^2 + 2\sigma^2 = 2r + \sigma^2$$